

MILOŠ STANOJEVIĆ

ASSOCIATE PROFESSOR AT UCL

SENIOR RESEARCH SCIENTIST AT GOOGLE DEEPMIND

RESEARCH INTERESTS

- Linguistics: Language Universals, Syntactic Formalisms and Incremental Parsing
- Learning: Neural Latent Variable Models and Probabilistic Inference over Structures
- Applications: Machine Translation, Sign Language Generation and LLMs in general

WORK EXPERIENCE

- Sep 2024 – now **Associate Professor in Computational Linguistics**
University College London
Linguistics Department, London, UK
- Nov 2021 – now **Senior Research Scientist**
Google DeepMind
Language Team, London, UK
- Nov 2017 –
Oct 2021 **PostDoc / Research Associate**
University of Edinburgh
School of Informatics, Edinburgh, UK
- July 2017 –
Oct 2017 **Internship**
Nuance Communications (now part of Microsoft)
Natural Language and AI group, Sunnyvale, US

EDUCATION

- 2013 – 2017 **Ph.D. Natural Language Processing**
University of Amsterdam, Netherlands
- 2011 – 2013 **Mgr. Mathematical Linguistics**
Charles University in Prague, Czechia
- 2010 – 2011 **M.Sc. Human Language Science and Technology**
University of Malta, Malta
- 2006 – 2010 **B.Sc. Electrical Engineering**
University of Niš, Serbia
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PUBLICATIONS

- arXiv 2026** [Syntactic Belief Update as the Driver of Garden Path Processing Difficulty](#)
arXiv preprint
Alan Zhou*, **Miloš Stanojević***, John Hale
*equal contribution
- HSP 2025** [Generalizing active gap-filling](#)
Conference on Human Sentence Processing
Tim Hunter, **Miloš Stanojević**, Edward Stabler
- COLING 2025** [Extracting structure from an LLM - how to improve on surprisal-based models of Human Language Processing](#)
International Conference on Computational Linguistics
Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny
- EMNLP 2024** [Do LLMs learn a true syntactic universal?](#)
Empirical Methods in Natural Language Processing
John Hale and **Miloš Stanojević**
- ACL 2024** [Multipath parsing in the brain](#)
Annual Meeting of the Association for Computational Linguistics
Berta Franzluebbers, Donald Dunagan, **Miloš Stanojević**, Jan Buys, John Hale
- CMCL 2024** [How can large language models become more human?](#)
best paper award *Workshop on Cognitive Modeling and Computational Linguistics*
Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny
- EMNLP System Demo 2023** [SynJax: Structured Probability Distributions for JAX](#)
Empirical Methods in Natural Language Processing System Demonstration
Miloš Stanojević and Laurent Sartran
- Cognitive Science journal 2023** [Modeling structure-building in the brain with CCG parsing and large language models](#)
top cited *Cognitive Science — journal*
Cognitive Science paper in 2023 **Miloš Stanojević**, Jonathan Brennan, Donald Dunagan, Mark Steedman, John Hale
- Neurobiology of Language journal 2023** [Neural correlates of long-distance dependencies in Chinese and English](#)
Neurobiology of Language — journal
Donald Dunagan, **Miloš Stanojević**, Maximin Coavoux, Shulin Zhang, Shohini Bhattachali, Jixing Li, Jonathan Brennan, John Hale

- EMNLP 2022 [Unbiased and Efficient Sampling of Dependency Trees](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević
- TACL
journal 2022 [Transformer Grammars: Augmenting Transformer Language Models with Syntactic Inductive Biases at Scale](#)
Transactions of the Association for Computational Linguistics — journal
Laurent Sartran, Samuel Barrett, Adhiguna Kuncoro,
Miloš Stanojević, Phil Blunsom, Chris Dyer
- Computational
Linguistics
journal 2021 [Formal Basis of a Language Universal](#)
Computational Linguistics — journal
Miloš Stanojević and Mark Steedman
- EMNLP 2021 [A Root of a Problem: Optimizing Single-Root Dependency Parsing](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Shay B. Cohen
- IWCS 2021 [Computing All Quantifier Scopes with CCG](#)
International Conference on Computational Semantics
Miloš Stanojević and Mark Steedman
- CMCL 2021 [Modeling incremental language comprehension in the brain with Combinatory Categorical Grammar](#)
Workshop on Cognitive Modeling and Computational Linguistics
Miloš Stanojević, Shohini Bhattachali, Donald Dunagan, Luca Campanelli, Mark Steedman, Jonathan Brennan, John Hale
- CASE 2021 [Modality and Negation in Event Extraction](#)
Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text
Sander Bijl de Vroe, Liane Guillou, **Miloš Stanojević**, Nick McKenna and Mark Steedman
- ACL 2020 [Max-Margin Incremental CCG Parsing](#)
Annual Meeting of the Association for Computational Linguistics
Miloš Stanojević and Mark Steedman
- IWPT 2020 [Span-Based LCFRS-2 Parsing](#)
The 16th International Conference on Parsing Technologies
Miloš Stanojević and Mark Steedman
- CUNY 2020 [Predictive Processing of Coordination in CCG](#)
Conference on Human Sentence Processing
Miloš Stanojević, John Hale and Mark Steedman

- ACL 2019** [Wide-Coverage Neural A* Parsing for Minimalist Grammars](#)
Annual Meeting of the Association for Computational Linguistics
 John Torr, **Miloš Stanojević**, Mark Steedman and Shay B. Cohen
- FG 2019** [On the Computational Complexity of Head Movement and Affix Hopping](#)
Conference on Formal Grammar
Miloš Stanojević
- NAACL 2019** [CCG Parsing Algorithm with Incremental Tree Rotation](#)
Conference of the North American Chapter of the Association for Computational Linguistics
Miloš Stanojević and Mark Steedman
- CMCL 2019** [The active-filler strategy in a move-eager left-corner Minimalist Grammar parser](#)
Workshop on Cognitive Modeling and Computational Linguistics
 Tim Hunter, **Miloš Stanojević** and Edward Stabler
- CogACLL 2018** [A Sound and Complete Left-Corner Parsing for Minimalist Grammars](#)
Workshop on Cognitive Aspects of Computational Language Learning and Processing
Miloš Stanojević and Edward Stabler
- PROPOR 2018** [SICK-BR: a Portuguese corpus for inference](#)
International Conference on Computational Processing of the Portuguese Language
 Livy Real, Ana Rodrigues, Andressa Vieira e Silva, Beatriz Albiero, Bruna Thalenberg, Bruno Guide, Cindy Silva, Guilherme de Oliveira Lima, Igor CS Câmara, **Miloš Stanojević**, Rodrigo Souza, Valeria de Paiva
- PhD Thesis 2017** [Permutation Forests for Modeling Word Order in Machine Translation](#)
University of Amsterdam 2017
Miloš Stanojević
- EMNLP 2017** [Neural Discontinuous Constituency Parsing](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Raquel G. Alhama
- ACL 2017 short paper** [Alternative Objective Functions for Training MT Evaluation Metrics](#)
Annual Meeting of the Association for Computational Linguistics
Miloš Stanojević and Khalil Sima'an

- LACL 2016 [Minimalist Grammar Transition-Based Parsing](#)
Logical Aspects of Computational Linguistics
Miloš Stanojević
- COLING 2016 [Hierarchical Permutation Complexity for Word Order Evaluation](#)
International Conference on Computational Linguistics
Miloš Stanojević and Khalil Sima'an
- COLING 2016 [Universal Reordering via Linguistic Typology](#)
International Conference on Computational Linguistics
Joachim Daiber, **Miloš Stanojević** and Khalil Sima'an
- WMT 2016 [Examining the Relationship between Preordering and Word Order Freedom in Machine Translation](#)
Conference on Machine Translation
Joachim Daiber, **Miloš Stanojević**, Wilker Aziz and Khalil Sima'an
- WMT 2016 [Results of the WMT16 Metrics Shared Task](#)
Conference on Machine Translation
Ondřej Bojar, Yvette Graham, Amir Kamran and **Miloš Stanojević**
- WMT 2016 [Results of the WMT16 Tuning Shared Task](#)
Conference on Machine Translation
Bushra Jawaid, Amir Kamran, **Miloš Stanojević** and Ondřej Bojar
- EMNLP 2015 [Reordering Grammar Induction](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Khalil Sima'an
- PBML 2015 [Evaluating MT systems with BEER](#)
The Prague Bulletin of Mathematical Linguistics
Miloš Stanojević and Khalil Sima'an
- WMT 2015 [BEER 1.1: ILLC UvA submission to metrics and tuning task](#)
Workshop on Machine Translation
Miloš Stanojević and Khalil Sima'an
- WMT 2015 [Results of the WMT15 Metrics Shared Task](#)
Workshop on Machine Translation
Miloš Stanojević, Amir Kamran, Philipp Koehn and Ondřej Bojar
- WMT 2015 [Results of the WMT15 Tuning Shared Task](#)
Workshop on Machine Translation
Miloš Stanojević, Amir Kamran and Ondřej Bojar

- SSST 2014** [Evaluating Word Order Recursively over Permutation-Forests](#)
Workshop on Syntax, Semantics and Structure in Statistical Translation
Miloš Stanojević and Khalil Sima'an
- WMT 2014** [BEER: BEtter Evaluation as Ranking](#)
Workshop on Machine Translation
Miloš Stanojević and Khalil Sima'an
- EMNLP 2014 short paper** [Fitting Sentence Level Translation Evaluation with Many Dense Features](#)
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Khalil Sima'an
- WMT 2012** [Selecting data for English-to-Czech machine translation](#)
Workshop on Machine Translation
Aleš Tamchyna, Petra Galuščáková, Amir Kamran, **Miloš Stanojević**, Ondřej Bojar

CONTRIBUTIONS TO THE RESEARCH COMMUNITY

- ARR Area Chair:** - ACL, NAACL, EMNLP, EACL, CoNLL conferences
- Standing Reviewer:** - TACL journal (Impact Factor 6.9)
- Computational Linguistics journal (Impact Factor 5.3)
- Reviewer:** - Natural Language Processing Journal (NLP) (Impact Factor 1.9)
- Language Resources and Evaluation Journal (LREC) (Impact Factor 1.8)
- Journal of Language Modeling (Impact Factor not available)
- Shared tasks organization:** - Workshop on Machine Translation (WMT) - Metrics task in 2015 & 2016
- Workshop on Machine Translation (WMT) - Tuning task in 2015 & 2016

SUPERVISION AND MENTORSHIP

Johns Hopkins University 2026- **Alan Zhou** (part of PhD dissertation committee)

UCL supervision 2026- **Francesco Cipriani** (PhD 2nd co-supervisor)
+ 8 MA students in Computational Linguistics specialization
+ 2 BA student in General Linguistics

UCL mentorship 2026 **Tal Nahari**
now Postdoctoral Fellow, Affective Brain Lab, UCL

2025 **Maximilian Moser**

now Research at King's College London

**Google DeepMind
internship mentor of PhD
students**

2026 **Yifu Qiu**
now PhD at University of Edinburgh

2025 **Zifan Jiang**
now PhD at University of Zurich

2024 **Matthias Lindemann**
now PostDoc at Instituto de Telecomunicações, Lisbon, Portugal

**Google DeepMind
scholar mentor**

2026 **Sindhu B. Hegde**
now PhD student at University of Oxford

2023 **Catherine Nana Nyaah Essuman**
now PhD student at Ontario Tech University

2022 **Mukhtar Mohamed**
now intern at Google DeepMind

**University of Edinburgh
supervision**

2019 **John Torr** (PhD Dissertation 2nd supervisor in the last year)
now RS at Apple Research, Cambridge

2021 **Seth Aycock** (MSc Dissertation main supervisor)
now PhD student at University of Amsterdam

2019 **Chiehmin Wei** (BA project supervisor)
now software engineer at Roblox

TEACHING

2025/2026 UCL	Lectures on Language Modeling within PLIN0072 Advanced Computational Linguistics
2025/2026 UCL	Lectures on Parsing within PLIN0034 Introduction to Computational Linguistics
2024/2025 UCL	Lectures on Parsing within PLIN0034 Introduction to Computational Linguistics
2024/2025 UCL	Present linguistics students with career opportunities in the industry UCL LingSoc Careers Event
2016/2017 University of Amsterdam	Natural Language Processing 2
2015/2016 University of Amsterdam	Natural Language Processing 2
2014/2015	short course on Monte Carlo Inference for PCFG

University of Amsterdam

2013/2014

University of Amsterdam

Elements of Language Processing

2012/2013

University of Amsterdam

Statistical Structure in NLP

INVITED TALKS

LSSC keynote speaker
Novi Sad, Serbia, 2026

Interaction of Linguistics and Computer Science Then and Now
Keynote speaker at conference *Language & Social Sciences in Contact: Artificial Intelligence (AI)*

FlaNN Seminar
Online, 2026

Modeling Sentence Processing Difficulty with Bayesian Syntax
Formal Languages and Neural Networks Seminar

Stony Brook University
New York, USA, 2025

Incremental Sentence Processing of Right-Adjunction

ILFC Seminar – CNRS,
Paris, France, 2025

Linguistic Universals in Grammars and Language Models
ILFC Seminar: Interactions between Formal and Computational Linguistics

Johns Hopkins University
Baltimore, USA, 2024

Syntactic Universals in Grammars and LLMs

University of Amsterdam
Netherlands, 2023

Syntactic Structures and Large Language Models

University of Cambridge
Cambridge, UK, 2023

The role of syntax in the world of large language models

UCL
London, UK, 2023

What is the relation of Syntactic Parsing, Language Universals, Brain Activity and Large Language Models?

ETH Zürich
Zürich, Switzerland, 2022

Testing Syntactic Theory on all Marr’s Levels of Analysis

Mathematics of Language
Montpellier, France, 2021

CCG Combinators in Brain Activity and Linguistic Universals

DeepMind
London, UK, 2021

Integrating Deep Learning with Deep Syntax

Nuance NL & AI lab
Sunnyvale, USA, 2017

Learning Reordering Grammars

Rich Parsing Workshop
Amsterdam, 2016

Minimalist Grammar Transition-Based Parsing

MT Marathon
Prague, Czechia, 2015

Discriminative Training for Statistical Machine Translation

GRANTS

- 2026 **Capital Equipment Investment Fund by UKRI**
£185,025 for computing equipment
Miloš Stanojević (lead applicant) and Mario Giulianelli (co-applicant)
- 2025 **Language Sciences Incubator Fund by University of Cambridge**
£5,000 for research project “Improving Language Equity and Inclusion through AI”
lead applicants: Songqiao Xie and Xinxin Yan
co-applicants: Mario Giulianelli, Napoleon Katsos, Miloš Stanojević

SCHOLARSHIPS

- 2010 – 2012 **Erasmus Mundus scholarship for EM-LCT Masters Program**
- 2004 – 2010 **Scholarship from the Ministry of Science and Technological Development of Serbia**
awarded for taking part in international mathematical competitions
- 2002 – 2004 **Scholarship from the Ministry of Education of Serbia**
awarded for taking part in international mathematical competition
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